



UNIVERSITY OF TRIESTE

Department of Mathematics, Informatics and Geosciences

Bachelor's Degree in Artificial Intelligence and Data Analytics

BACHELOR'S THESIS

**Cross-View Association of Identical Parts Across
Non-Overlapping 3D Scans: Identifiability,
Shift Ambiguity and Evaluation**

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Academic Year 2025/2026

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Chapter 1

Introduction

1.1 The problem

Industrial handling is increasingly automated. Repetitive and heavy tasks such as lifting or sorting parts, once assigned to human operators, are now delegated to robots to satisfy the cycle time required by the production line.

For a robot to grasp an object, it is necessary to determine the object's pose, which dedicated sensors measure as a position and an orientation relative to the robot's own coordinate frame.

For a single compact piece, one 3D sensor is usually sufficient, as its viewpoint can cover the entire object. On the other hand, a very long part cannot be captured from a single viewpoint, making a single scanner insufficient. The layout therefore requires two sensors, one for each end. The part is gripped precisely at these ends, whose faces also determine the 6D pose of a straight extruded profile up to the symmetry of the cross-section.

The setup consists of two structured-light 3D scanners, positioned at either end of a conveyor carrying bundles of six-metre metal bars of identical profile.

Each device is mounted on a dedicated carriage at one end of the bundle, facing the end faces of the bars rather than their length. The two carriages are separated by several metres along the axis of the bar. The part of the bundle that each scanner covers is often narrower than the bundle itself, so a single view does not always contain every bar. Because the two sensors face opposite ends, their respective viewpoints neither overlap nor capture any common surface. Finally, a dual-arm manipulator picks up each bar by gripping it at both ends. Figure 1.1 illustrates this setup.

The task is therefore to determine which detection from one scanner and which from the other belong to the same physical object. In this specific case, data association cannot be resolved using conventional techniques due to the absence of typical geometric or visual features.

Absence of epipolar constraint. The lack of overlap between the two views means that the scanners never see the same surface.

Visual appearance invariance. Because the bars originate from the same extrusion batch, end faces belonging to the same profile family (i.e., bars sharing the same cross-section) are identical up to saw marks.

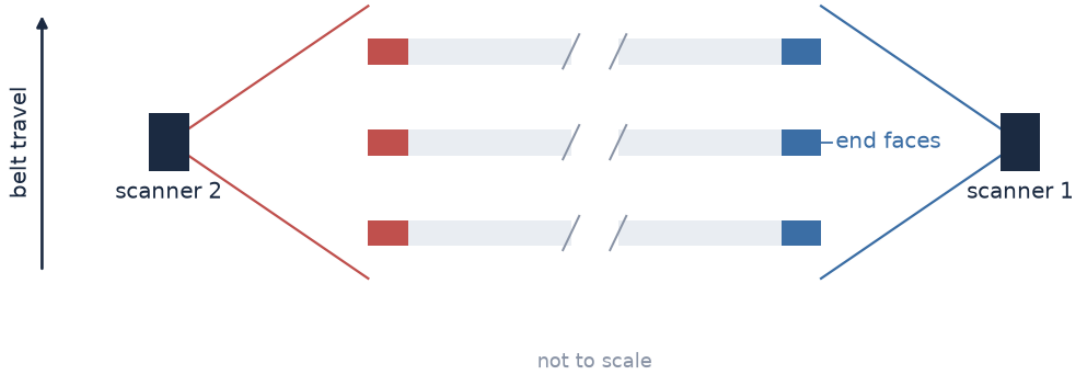


Figure 1.1: Dual-scanner system. Top view of the system. Each scanner is positioned at one end of the bundle and detects only the end faces of the bars facing it. In the figure, these faces are highlighted at both ends of each bar. The viewpoints of the two scanners do not share any surface and the bundle passes through both. Neither axis is to scale.

Relative position invariance. The bars are packed edge to edge at an almost uniform pitch, so the scanner detects a sequence of end faces that remains visually invariant even if the bundle shifts by one position.

Conveyor motion between acquisitions. Because the conveyor belt advances between the two acquisitions, the two scan windows can cut the bundle at different bars, in which case an association that pairs the detections in their order is shifted by one or more bars.

This shift ambiguity is illustrated in fig. 1.2. In each view, a different end bar remains excluded because each view often covers less than the whole bundle and the bundle moved between the two acquisitions. A correct association should record two unpaired detections, whereas a complete sequential association would lead to a systematic error.

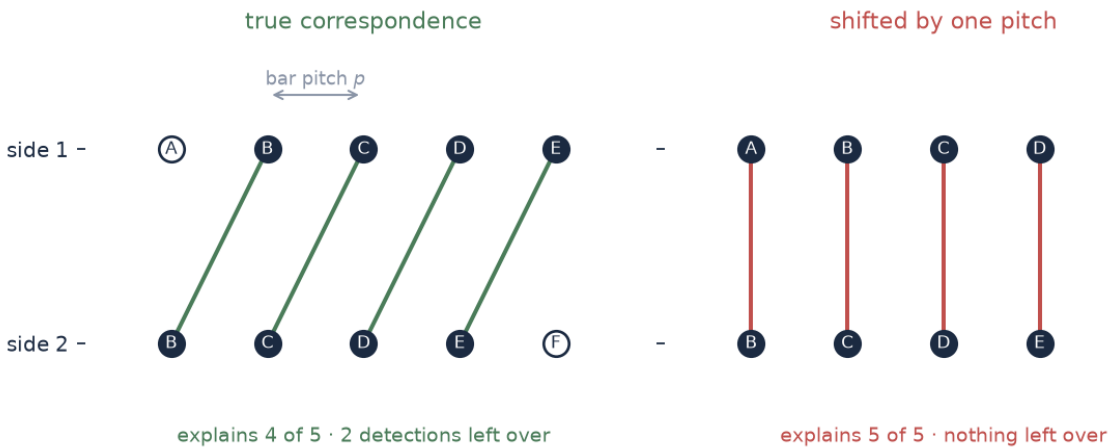


Figure 1.2: The shift ambiguity (schematic). Each scanner covers a different part of the bundle, so a different bar is truncated in each acquisition. A correspondence shifted by one pitch would incorrectly pair all detections, whereas the correct correspondence leaves the two truncated detections unpaired. Letters identify the physical bar and hollow circles mark the unpaired detections.

This condition is not an edge case because in many scenes of the dataset some bars are

scanned at only one end. The objective is to determine what portion of the correspondence is actually recoverable and to define how unpaired detections are handled.

The scans used in this thesis were provided by beanTech, the company that developed the vision software of the rig for the manufacturer that operates it. At the time of this work the rig was still being commissioned and only a limited number of scenes was available. This setting favours methods that require no training data, such as the classical detector and the pairing methods studied here, since the detector needs only the drawing of the bar cross-section.

1.2 Related work

To the best of our knowledge, no method in the literature addresses this specific case. Cross-view association is a widely studied problem, but existing approaches rely on assumptions that do not hold in this setting, such as the presence of a distinctive visual appearance or the geometric overlap of fields of view. Four main lines of research can nevertheless be identified, each resting on assumptions whose absence in this setting has specific consequences.

Geometric correspondence and assignment. The first line of research assumes the presence of surfaces observed in common. Methods based on stereo matching and multi-view correspondence, starting from dynamic programming along scanlines Ohta et al. 1985 [9], function precisely because the same surface appears in multiple images. The same principle applies to systems that associate identical instances detected by different cameras using geometry alone. These systems calculate the epipolar distance for each pair of detections, model it as a graph cost and recover the groups by clustering Sun et al. 2025 [13]. In the absence of visual overlap, however, the epipolar constraint no longer applies. The task thus reduces to a pure assignment problem, in which the cost matrix must be derived directly from the scene layout. The Hungarian algorithm solves minimum-cost bipartite matching Kuhn 1955 [4] without imposing order constraints on correspondences. An assignment can also be constrained to preserve the order of the elements, which is an explicit modelling assumption and does not follow from epipolar geometry. The monotonic alignment compared in this thesis (section 2.4) imposes such a constraint on the order of the bars.

Discriminative appearance. The second line of research leverages visual appearance to discriminate one object from another. Benchmarks for cross-camera instance association, such as MessyTable Cai et al. 2020 [1], are designed for objects showing appearance differences between instances, even minimal ones. According to their analysis, reliable association requires appearance, context and geometry together. The application most closely related to the case under consideration is found in the timber industry, where images of the same log end face taken at different times are matched as a biometric fingerprint, either with texture features that describe the annual ring pattern Schraml et al. 2015 [11] or with features learned by a deep network Xie et al. 2025 [15]. This approach is effective because each end face presents a unique surface, unlike a metallic extrusion whose cross-section is geometrically and visually identical to adjacent ones.

Identifiability and observability. The third line of research concerns the information content of the data themselves. In the absence of overlap and characterising visual attributes, the task reduces to an unlabelled sensing problem Unnikrishnan et al. 2018 [14]: observations are known, but the correspondence matrix is unknown. The fundamental question lies in determining what can actually be inferred from them. The literature addresses the topic along two directions: defining which configurations are reconstructible and formulating a converse that bounds what no algorithm can recover. The same critical issue emerges

in self-calibration for system parameter estimation, where it is addressed by studying the observability of the information matrix Maye et al. 2016 [7] and identifying unobservable directions in the parameter space Nobre et al. 2019 [8].

Association performance evaluation. The fourth line of research concerns association evaluation, which requires metrics distinct from those used for pure detection quality. Several studies on multi-target multi-camera systems highlight that single-frame metrics do not provide information on identity consistency across space and time. For this reason, identity-focused precision and recall metrics were introduced Ristani et al. 2016 [10], followed later by the HOTA metric, which prevents an aggregate index from masking the source of error between detection and data association Luiten et al. 2021 [6]. The BOP benchmark applies an analogous principle to 6D pose estimation, defining correctness in relation to the symmetry group of the object rather than relative to a canonical pose Hodaň et al. 2020 [3]. Finally, concurrent work shows that pairwise ranking metrics and one-to-one assignment metrics can disagree, which means that a model may achieve a higher score in ranking metrics without a corresponding improvement in its actual assignment capability Shelukhan et al. 2026 [12].

The gap addressed by this work lies between identifiability and evaluation. For an acquisition setup without visual overlap that observes identical elements at a regular spacing, it has not yet been formalised which part of the matching problem can be resolved in principle and which part is intrinsically undetermined. Consequently, reading an overall accuracy metric requires separating correctable algorithmic errors from structural limitations of the physical setup.

1.3 Contributions

This thesis determines what the data of two non-overlapping views can recover. The geometry of the rig is recovered except for the absolute position of the scanners, whereas when the bars are evenly spaced the geometry of the scene determines the correspondence between the detections only up to a shift by a whole number of bars. The pairing methods are then evaluated on the basis of this distinction rather than through a single aggregate accuracy figure. The contributions of this work are the following.

1. **The mechanism of the shift ambiguity.** Besides the true association, the data are compatible with a family of associations shifted by a whole number of bars along the bundle. When the bars are evenly spaced, a shifted association has practically the same geometric cost as the true one, because the translation estimated for the scene changes by the same shift and the distances between paired faces stay the same. The two associations differ only in the number of detections left unpaired, which is lower for the true association when both views contain the same bars. For this reason the ambiguity depends on whether the scan window truncates the two views differently and not on how far the bundle moves between the two acquisitions (section 2.5 and ???).
2. **Confirmation on real data.** The errors of the most accurate pairing method on the real scenes follow this mechanism. Every scene on which the method fails completely is shifted as a whole and contains at least one detection without an annotated partner. In each of these scenes the wrong association also has a lower cost than the annotated one under the objective that the method minimises, so these errors depend on the information contained in the detections rather than on the search for the minimum (??).

3. **Identifiability of the rig geometry.** Prior knowledge of the nominal bar length determines the separation between the two carriages but not their absolute position, which the scene content does not constrain. Alternating calibration and assignment recovers the separation and removes an error of about one metre along the bar axis (sections 3.2 and 3.3). The length check of the original system cannot detect a shifted association, because pairing the ends of two adjacent bars changes the reconstructed length by far less than the tolerance of the check (section 4.1).
4. **The hybrid method.** The pairing method proposed in this thesis applies the order constraint of the monotonic alignment within each stacking level and admits a bar crossing its neighbours only when the detections support it (section 2.4). With both of the detectors compared in this thesis it is the most accurate of the methods tested, although on this dataset its advantage over pairing the detections by their order alone is not statistically significant (????).
5. **Evaluation of the pairing methods.** The counting metric checks only how many pairs are assigned and therefore does not distinguish a correct association from a shifted or swapped one. This thesis relies instead on a correspondence annotation linked to the detections by position, which makes it possible to score the output of any detector against the same ground truth (section 2.1 and ??).
6. **Development and audit of a synthetic twin.** The synthetic twin, developed to simulate scene acquisitions, replicates detection counts and detector accuracy but does not model how far the bar ends stagger along the bar axis. This restricts the scope of any validation conducted on it to the transverse plane alone (section 2.6).

Chapter 2

The data, the rig and the task

Apart from the controlled experiments on the synthetic twin (section 2.6), all results in this thesis come from real acquisitions. Every comparison is between pairing methods that operate on the same detections and return cross-view pairs. A detection is an end face identified in a single view, whereas a pairing method is the procedure that takes the detections from both views as input and returns a set of pairs connecting them.

2.1 The dataset, annotations and evaluation metrics

A scene consists of a bundle of bars acquired once from each end, yielding two point clouds. The dataset comprises 108 scenes and 216 clouds, classified by profile family (table 2.1), with each scene associated with a manually assigned scenario category.

Table 2.1: Profile families of the dataset. For each family, the shape of its cross-section as given by the CAD drawing, the number of scenes and the number of annotated faces. The L-profile appears twice because it was acquired in two separate campaigns, so the five families share four distinct cross-sections. The five families total 108 scenes and 935 faces.

profile family	cross-section	scenes	annotated faces
complex profile	complex	4	15
heavy profile	rectangular, 80×95 mm	11	68
L-profile	asymmetric L-shape	40	274
L-profile, second acquisition	asymmetric L-shape	24	223
square profile	square, 40×40 mm	29	355

The second column indicates the cross-section shape, obtained from the CAD drawing of the profile. Although the dataset consists of five families, there are only four distinct cross-sections because the L-profile appears twice due to two separate acquisition campaigns.

The categories describe the arrangement of the bars in the bundle: single (56 scenes), overlapping (22), standing (15), crossed (9) and tilted (6). Across all scenes, a single side features a median of 4 faces and a maximum of 29.

Truncation, where a bar is visible to only one scanner, affects 30 of the 108 scenes and leaves 49 of the 935 faces without a counterpart in the other view. Consequently, the assumption that every bar is visible to both scanners fails in roughly one out of four scenes.

Two types of annotations have been defined.

The first is a count annotation, indicating how many end faces are visible for each side. It was made first, by hand, on images generated from the point clouds and independently of any detector.

The second is a correspondence annotation that, for each scene, provides a list of claims: inter-view pairs, each equipped with an anchor point on both faces, taken from the centroid of the face detected by the candidate-pool detector and confirmed by the annotator. In total, this annotation contains 407 claims, with at least one in every scene. The anchors allow each claim to be associated with the detections of the detector being evaluated, verifying whether the detector successfully produced that pair. A claim is resolved for a detector when both of its anchors can be linked to a detection of that detector. Where a scene needs it, the annotation also carries a note recording, for example, which profile is truncated in one view or not confirmed by the detector.

The two annotations support different metrics. A counting metric checks only how many pairs a method produces, so the criterion is satisfied whenever the number of pairs is correct, even if all pairs are incorrect. For example, if two true pairs are swapped, the method still produces two pairs and the counting metric counts them as two true positives. The same happens when one view truncates a bar that the other contains, in which case an association shifted by one position along the row keeps the annotated pair count. Only the correspondence annotation indicates which pairs are correct and can therefore detect a swap. Work on multi-camera tracking faced a similar limitation and resolved it with identity-based precision and recall metrics Ristani et al. 2016 [10].

Four metrics are evaluated:

- **Counting precision, recall and F1:** compare the number of faces or pairs produced by a method against the count annotation, both scene-by-scene and aggregated. A true positive is defined as the smaller of the two numbers, calculated per side for faces and per scene for pairs.
- **Correspondence recall:** the proportion of claims correctly reproduced by a method (i.e., with correct bar identity), evaluated over the set of resolved claims. Restricting the score to resolved claims prevents a missing detection from being counted as a pairing error, following the distinction formalised by HOTA for tracking Luiten et al. 2021 [6].
- **IPAA-1.0** (Image-Pair Association Accuracy at 1.0): the proportion of scenes correctly resolved in their entirety (referred to as IPAA-100 in MessyTable Cai et al. 2020 [1]). While aggregated recall gives greater weight to scenes with many associations, IPAA-1.0 weighs every scene equally. Furthermore, it aligns most closely with a line operator’s experience, where an entire bundle is processed as a single unit.
- **Inversions:** the count of pairs within the same row that lie side-by-side at the same height in both views but exhibit an inverted Y-order between them. These capture the swaps that counting metrics cannot detect by design.

In this thesis, all confidence intervals are calculated by resampling bundles rather than pairs. Pairs within the same scene share the same bundle, transport estimate and assignment, so they are not independent samples and an error tends to affect the whole scene rather than individual pairs. Some scenes are also repeat captures of the same stationary bundle, whose point clouds coincide within 2 mm on almost every point. These scenes are grouped and resampled together, so the 108 scenes of the dataset form 96 independent units, called bundles

in the rest of the thesis. For correspondence metrics all of them are resampled, since each scene contains at least one claim.

All results reported here are based on the candidate-pool detector (section 2.3). The anchors were defined from its output, so every claim corresponds to one of its detections and all 407 claims can be scored.

The alternative detector, based on two-pass suppression (section 2.3), resolves 382 of the 407 claims and re-associates faces with a median error of 0.96 mm, compared to 0.00 mm for the candidate-pool detector. The comparison between the two detectors therefore measures how robust each method is to a change of detector rather than which detector is more accurate.

2.2 Acquisition geometry

Two scanners are placed at opposite ends of a conveyor belt, each mounted on its own linear carriage. The global reference frame (in millimetres) is defined relative to the robot: the X-axis aligns with the bars and carriage motion, the Y-axis indicates the conveyor feed direction and the Z-axis represents height. Calibration relates each scanner’s local frame to the global frame via a rigid transformation $p_{\text{world}} = R p_{\text{scanner}} + t$. After calibration, the point clouds are orthographically projected onto the YZ plane (feed and height), which lies nearly frontal to the end faces.

Two calibrations are available. The offline calibration, defined by a single rotation and translation knot per scanner, yields a constant rigid transformation valid exclusively at the carriage position where knots were acquired. By contrast, the original system calibration uses ten knots per scanner across roughly 2.1 m of travel, interpolating between them with a centripetal Catmull-Rom spline for translation and spherical cubic interpolation (SQUAD) for rotation, with values clamped at the boundaries. Applying the original calibration would require the carriage position of every scan, which is absent from the dataset, so the constant offline calibration was used instead. Consequently, accuracy is limited to the single acquired position, introducing an error along the X coordinate, the axis of carriage motion, that is subsequently measured and compensated (section 3.3). Figure 2.1 shows the two clouds of one scene in the world frame.

Several quantities in table 2.2, along with certain detection constants, originate from the source code of the original system installed on the machine.

2.3 Detection

The points that survive the radial filter are projected onto the YZ plane and a look-up table records which 3D points correspond to each pixel, enabling a 2D match to retrieve its 3D support. The profile cross-section is rasterised from the CAD drawing (DXF file) at 2.5 px/mm and slid over the image at 2° intervals across the full rotation. Every local cross-correlation maximum reaching the lost-face threshold of 0.2 constitutes a candidate (the threshold is defined relative to the template’s self-correlation peak).

The two detectors used in this work share this initial stage but differ in the criterion used to confirm candidates.

The two-pass suppression detector decides on the 2D image: it confirms a candidate when the peak reaches the match threshold of 0.4 and the mask overlap exceeds 0.65, suppresses

Table 2.2: Quantities of the rig and of the acquisitions. For each quantity, its value and how it was obtained. Lengths are in millimetres in the world frame, where X runs along the bars, Y along the conveyor feed and Z in height.

Quantity	Value	How it was obtained
Standoff	656 mm, interquartile range (IQR) [641, 669] mm	Median radial distance between the scanner origin and the retained points across 216 clouds.
Angle of incidence on the end face	22.0°	Extracted from the production calibration rotation matrices. Stable within $\pm 0.35^\circ$ over roughly 2.1 m of carriage travel.
Angular envelope	$[-19.3^\circ, +17.4^\circ]$, union of both sides	Field of view width of each scanner along its tangential direction, within its local frame. The lower limit repeats within 0.06° across 104 of 108 scenes on both sides, indicating a hardware limitation.
Scan window	median 258 mm, maximum 414 mm	Calculated between the 0.5th and 99.5th percentiles of points retained after the 850 mm radial filter. Prior to filtering, the median extent is 479 mm. The effective window depends jointly on this radial filter and the device’s angular span.
Bar pitch	[38.6, 118.4] mm	Range of median distances between adjacent bars, calculated for each of the five families on the correspondence anchors.
Cross-view transport	median $ \Delta Y $ 32.1 mm	Calculated per scene as the absolute value of the median ΔY between the anchor points of the two views. The reported value represents the median of these values across all 108 scenes.
Nominal bar length	6005 mm	Value set by the operator in the original system, independent of the profile family.
Distance between carriages	5403.3 mm	Estimated via self-calibration (section 3.3) rather than direct measurement. It differs by 1.3 mm from the documented separation of 2701 mm per side (5402 mm total).
Effective resolution	2.5 px/mm	Product of the rendering scale (10 px/mm) and the downscaling factor (0.25).

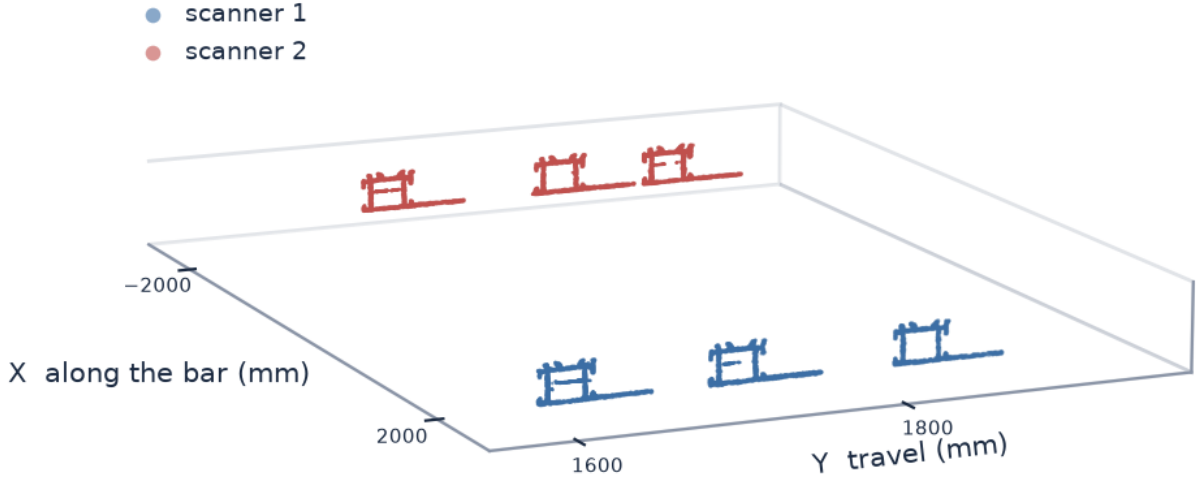


Figure 2.1: Two views of one scene in the world frame. The two point clouds of a single scene in the global reference frame. The radial filter, applied before calibration, discards any point beyond 850 mm from its scanner, which isolates the end faces from the machine and the conveyor belt and leaves 15,037 of the 298,061 points on the first side. Nothing is ever acquired between the two clouds, so the bar is never fully scanned. The scene belongs to the L-profile family.

duplicates at an intersection over union (IoU) of 0.2, clears the confirmed footprints from the image and repeats the sweep so that a hidden face behind a stronger neighbour can emerge in the second pass.

The candidate-pool detector instead decides based on the 3D points for which candidates compete. It retains every candidate above 0.2 without angle limits or bounding box overlap suppression, assigning to each candidate the 3D points within its footprint at its depth. Finally, a single global selection picks the subset explaining the most points against a fixed per-instance cost, with the constraint that no two selected candidates claim the same point. The candidate-pool detector is the one used in the remainder of this work, while two-pass suppression was retained to repeat the comparison between pairing methods on a different underlying detector. Figure 2.2 shows how the two detectors interpret the same image.

For each detected face, both methods return the 3D footprint points by masking the crop with the template shape and tracing back through the look-up table. Extracting the centroid and normal of the fitted plane via total least squares yields inputs for the pairing gates, combined with the matching angle retained from the 2° quantised sweep.

In this detection phase, the resolution, the 2° step size and the correlation thresholds come from the original system installed with the machine. After identifying a face and retrieving its 3D points, that system executed an extra step: it re-projected the face onto its own plane, estimated using RANSAC Fischler et al. 1981 [2], and repeated the angle search on that view with a finer step. It thus obtained a refined angle and a planar reference for each face. The detection pipeline used in this thesis does not perform this step. A more precise angle would tighten the consistency checks between the ends of the same bar without helping to decide which end belongs to which bar, since identical parallel bars share the same angle.

Across all 108 scenes, the candidate-pool detector achieves a precision of 0.986, a recall of 0.955 and an F1 score of 0.970 against per-side counts. A subsequent recovery stage increases recall to 0.967 and F1 to 0.976. Operating on unpaired faces, this stage estimates

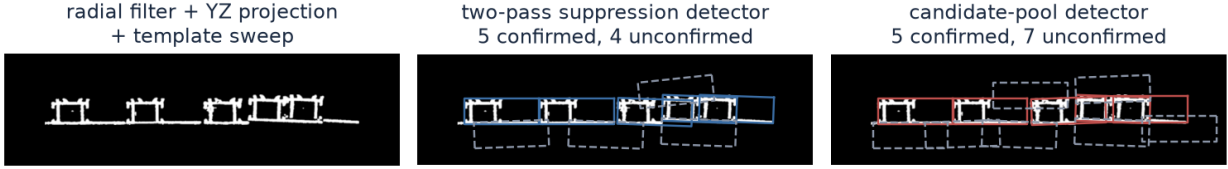


Figure 2.2: Detection. On the left, the projection on which both detectors operate after the radial filter and the template sweep. In the centre and on the right, the interpretation of the same image by each detector: confirmed detections are enclosed in coloured bounding boxes, while unconfirmed candidates appear in dashed grey boxes. In this case, both confirm the same five faces, but differ in the amount of residual unconfirmed evidence: four candidates versus seven.

their position in the opposing view using the cross-view translation, searches for a local match and accepts it only if the 3D footprint exhibits valid end-face characteristics. Approximately 20% of attempts are accepted.

As a baseline for the classical detector, a small convolutional network was trained on the same annotations and with the same code. It predicts face centres as peaks of a Gaussian heatmap, with a second head recovering their offset within the output cell. This formulation follows the centre-point detection proposed in Zhou et al. 2019 [16], whose Gaussian target and loss function derive from Law et al. 2018 [5]. Detection labels give only face centres, so a detector predicting bounding box extent and orientation would lack data to learn those parameters. Direct centre prediction instead keeps both model branches focused on per-side counts and the position of each face.

The results of the learned model depend on how the data is split. In the scene-disjoint split, where the training and test scenes differ but contain the same cross-sections, the model achieves a test F1 of 0.902. In the family-disjoint split, the test family is never seen during training and the two L-profile acquisitions are held out together, yielding an F1 of 0.425. The drop from 0.902 to 0.425 shows that the model recognises previously seen sections well, but fails to generalise to novel geometry. Both values are averages across two training runs, in each of which the model sees only a subset of the annotations, and should be compared with the classical detector’s 0.970. A paired bootstrap over bundles on one of the two runs gives, in both splits, a confidence interval for the difference that lies entirely below zero, so the learned model is significantly less accurate than the classical detector.

The discrepancy between the two splits shows that the model learns the four cross-sections present in the dataset, but does not develop a general end-face detector. Given a DXF drawing, the classical detector requires no training data and therefore always operates in the family-disjoint regime. The price is inference time, which is 2.53 s per scene side against 18 to 24 ms for the learned model. The two F1 values are lower bounds, because training was stopped before the test F1 had converged under a CPU-only schedule. The qualitative conclusion does not depend on them, since the model does not generalise to an unseen cross-section.

2.4 The association task and the methods compared

The association task starts from the faces detected in the two views and determines which faces, one per view, represent the two ends of the same bar. The inputs are therefore the detections from the two views and the output is a set of pairs between detections. Figure 2.3

shows an example.

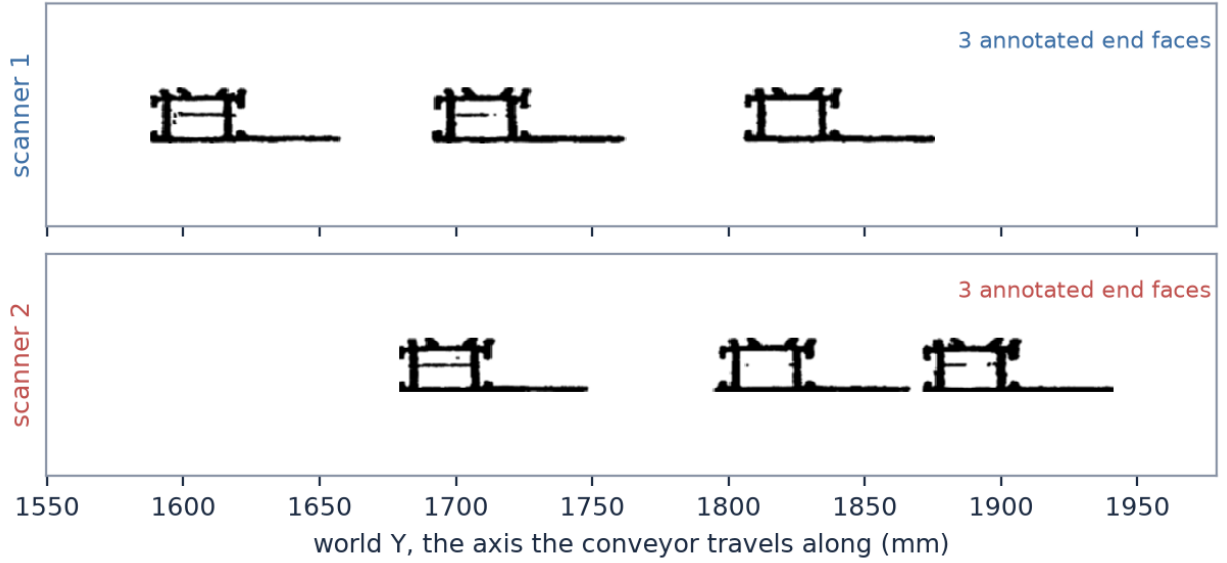


Figure 2.3: The same bundle seen from the two ends. YZ projection on which the two detectors operate for the same scene. The two views are drawn on the same Y-axis of the world reference frame, making the face positions directly comparable. The faces are identical up to the saw marks but the two views do not coincide along Y because the conveyor belt moves between the two acquisitions. The scene belongs to the L-profile family.

The faces detected in a view are grouped into stacking levels, i.e., the horizontal rows of the bundle. A face is one level above another when the Y coordinate of its centroid falls within the Y interval of the first and its centroid's Z coordinate exceeds the maximum Z of the second. This relationship is iterated until stabilisation and the faces in each level are sorted by decreasing Y. The grouping is relative to the bundle because no belt plane measurement is available and the original system provides no nominal value for it.

The task is a partial association between the detections in the two views. Each detection includes a YZ centroid, a coarse matching angle, the estimated face normal and the stacking level to which it belongs. Given the detections $D_1 = \{d_1, \dots, d_m\}$ and $D_2 = \{e_1, \dots, e_n\}$ of the two sides, an association is a partial matching $\pi \subseteq D_1 \times D_2$ in which each detection appears at most once. Unpaired detections are allowed and, in this dataset, necessary because 49 faces have no counterpart.

All association methods compared in this thesis share a single cost model:

$$c(i, j) = \frac{|\Delta\theta(i, j)|}{\tau_\theta} + \min\left(\frac{\|\Delta_{YZ}(i, j)\|}{\tau_d}, \kappa\right) + \lambda_L |\text{level}(i) - \text{level}(j)| + \lambda_Z [\text{outlier}(i) \neq \text{outlier}(j)] \quad (2.1)$$

The cost consists of four terms. The first term, $\Delta\theta$, measures the orientation difference between the two faces. The cross-section can be symmetric, so two orientations differing by a full period are considered equal: 90° for a square section, 180° for a rectangular section and 360° for an asymmetric section. The period is measured from the section contour and is not declared per family, in line with symmetry handling in BOP 6D benchmarks Hodañ et al. 2020 [3]. Figure 2.4 shows the four cross-sections with their corresponding measured periods.

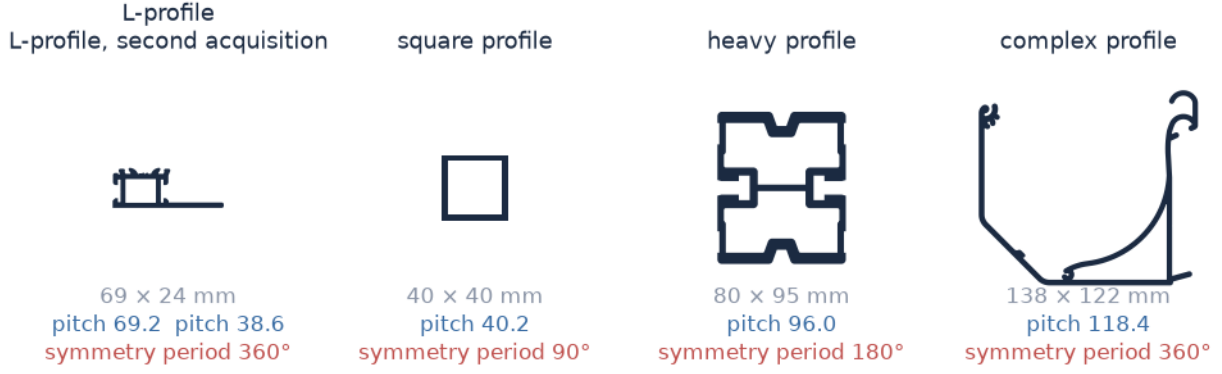


Figure 2.4: The four cross-sections. The five families of the dataset include four distinct cross-sections, as the L-profile was acquired twice. Each section is read from the technical drawing DXF through the same loader used by the detector’s template renderer. For each section, the bounding box, the bar pitch measured for each family using it and the rotational symmetry period adopted by the cost model are shown. The L-profile panel shows two pitch values, one per acquisition. The square section has a 90° period, the rectangular section 180° , while the L-shaped and complex sections have a 360° period.

The second term, Δ_{YZ} , is the distance between the centroids in the projection plane, saturated at κ . Saturation is necessary because the translation between the views is estimated and never exact, so the distance must remain a soft prior that still allows a transverse bar to be paired.

The third term penalises pairs belonging to different stacking levels. The fourth term penalises pairs in which one face is a height outlier and the other is not, where a face counts as an outlier when its centroid Z sits more than 8 mm from the median height of its own view. This term is kept separate from the distance term because it still discriminates after the distance has saturated. The scaling factors make a cost of 1 equivalent to the tolerance threshold: $\tau_\theta = 2.5^\circ$, $\tau_d = 45$ mm, $\lambda_L = \lambda_Z = 2$.

Two constants vary depending on the method. A method that does not estimate the cross-view translation uses $\kappa = 3$ and keeps a pair only when its cost is below 6, the cost of leaving both of its faces unpaired. Conversely, methods that estimate the translation increase these values to 4.5 and 7.5. This is because in methods compensating for the transport, a true pair has a small but non-zero residual due to detection noise. Consequently, the saturation threshold can be higher, allowing a legitimate pair with a large residual to still be associated. However, the cost of leaving two faces unpaired must remain above κ , otherwise it would be preferable to leave everything unpaired. In this way, a true pair with a large residual costs less than leaving its two faces unpaired. All costs cited in this thesis are based on methods that estimate the translation.

Seven pairing methods are compared in the rest of this work.

- **Order-based anchors.** Associates the i -th face on one side with the i -th on the other within each stacking level, without estimating any cross-view translation. After matching, this method applies the original system’s aggregation gate, which accepts a pair only if it passes three checks: torsion within 2.5° , nominal length within ± 10 mm of 6005 mm and tilt within 2.5° . However, since offline calibration introduces the bias on world X described in section 2.2, the length check rejects every pair and the method yields zero matches. To isolate the effect of that check, three variants are

defined. The first removes the length check and is taken here as representative of the installed heuristic, since in the field the threshold is configured to 2000 mm instead of ± 10 mm. The second keeps the check but centres its tolerance on the estimated bias. The third applies none of the three checks and keeps only the order of the detections, which separates what the heuristic owes to its ordering from what it owes to its checks. All three variants rely only on the order produced by the detector, without estimating anything, which is why they are termed order-based anchors.

- **Free assignment.** Solves the assignment problem on the shared cost model without estimating any translation between the views Kuhn 1955 [4]. A constant penalty for unpaired faces is added to the cost matrix, allowing some instances to remain unpaired.
- **Aligned assignment.** Estimates a YZ translation between views for each scene via RANSAC consensus Fischler et al. 1981 [2], in which every pair of centroids proposes a translation and the one with the most inliers is selected. The assignment is then solved on the translated centroids and the translation is refitted once as the median residual of the matched pairs.
- **Monotonic alignment.** Introduces an order constraint. The method assumes that the relative order of the bars is preserved between the two views, so that a bar to the left of another in one view is also to its left in the other. Under the same translation estimate, both sides are sorted along Y and the assignment becomes a sequence alignment problem solved via dynamic programming Ohta et al. 1985 [9], with the order constraint imposed directly on the assignment. The constraint is enforced across the entire bundle rather than level by level, which makes it incorrect when a bar rests on top of others.
- **Hybrid method.** This is the approach proposed in this work. It applies monotonic alignment within each stacking level. The two views do not always report the same number of levels, for example when the top bar of a tall stack is recognised as a separate level in one view only, so in that case the levels are matched by their median height rather than by their index. A single ordered assignment over the entire bundle is used only when one of the views has a single level. It adds an explicit hypothesis for bar crossovers, which is accepted in two cases. In the first, both views independently observe the same number of bars raised relative to the rest of the bundle and those bars break the rigid motion of the bundle between the two views. In the second, a bar lying across the bundle rests on its neighbours at one end and on the bed at the other, so it is raised in one view only and has no counterpart near it once the translation between the views is applied. That bar is paired with the faces left unpaired on the other side using only orientation and distance, because the level and height terms of the cost model penalise precisely the elevation that identifies it.

2.5 The shift family and identifiability

The bars of a stacking level lie next to each other in a fixed order from left to right, with an almost constant pitch p within the bundle. The visible faces of a level are indexed along Y on each side, with the indices of side 2 offset so that the true correspondence pairs face i of side 1 with face i of side 2. For an integer k , the shift family is defined as

$$\pi_k = \{ (i, i + k) : \text{both indices exist within the level} \} \quad (2.2)$$

which associates face i of side 1 with face $i + k$ of side 2, leaving the $|k|$ faces at the boundary of each view unpaired. The true correspondence is therefore π_0 . Each π_k is a valid partial correspondence (section 2.4), so the shared cost model assigns a cost to each of them.

The shift family is important for three reasons:

Geometric degeneracy under exact regularity. When the pitch is constant, shifting the association by k positions and increasing the estimated translation by $k \cdot p$ leaves the YZ distances between paired faces unchanged. For a method that estimates a translation for each scene, the geometric terms of the cost therefore take the same value for π_k and for π_0 . This is the degenerate case of the unlabelled sensing problem Unnikrishnan et al. 2018 [14], in which the correspondence can be recovered only from generic measurements, so that a bundle with a constant pitch is the extreme case of non-generic measurements.

Unpaired detections and truncation. Since the bundle is finite, π_k leaves unpaired the $|k|$ faces at the boundary of each view, whereas π_0 pairs every bar that both views contain. When the two views contain the same bars, the true correspondence therefore leaves fewer detections unpaired than any shifted one, whatever the transport. Pitch irregularity adds a geometric difference between the two, because an irregular bundle shifted by a whole pitch no longer matches the other view. Truncation that differs between the two views has the opposite effect. When one view cuts off a bar that the other contains, the true correspondence also leaves detections unpaired, so that against a shift towards the truncated side only the geometric difference remains, which is small when the pitch is regular. Neither the regularity of the pitch nor the position of the bundle in the scan window is under the control of the vision system.

The family is a proper subset of possible errors. A repermutation (i.e., a permutation that alters the relative order) corresponds to a bar crossing over its neighbours. No π_k contains a correspondence of this type. On real scenes, and restricted to the detections that have an annotated partner, the only associations that the cost model ranks above the ground truth leave a crossing bar unpaired, so they also fall outside the family (??). On the full detection set, by contrast, the wrong outputs of the hybrid method are shifts that cost less than the annotated pairs (??).

2.6 The synthetic twin and its limits

The synthetic twin is a geometric simulator of the line. It generates bars by extruding the DXF cross-sections along the axis, arranges them into bundles with known 6D poses and simulates two virtual scanners by ray-casting onto the calibrated geometry. The output is saved in the same scene files used for real data, allowing the pipeline (loading, detection and pairing) to process them without modification. The bars are generated by the simulator, so their identity is known by construction and requires no annotation. This enables controlled experiments that the real dataset cannot offer: variations in bar pitch regularity, inter-view transport, bundle position and the cost model margin relative to an artificially constructed member of the shift family. Figure 2.5 compares a real and a simulated scene.

The audit is conducted on the 56 scenes of the single category. Detection counts per scene-side agree at a Kolmogorov–Smirnov statistic of $D = 0.04$ and detection accuracy is comparable, with precision and recall of 0.974 and 0.964 on real scenes against 1.000 and 0.918 on the twin.

Point statistics, however, do not match, since the twin contains 46% of the real cloud size and only 19% of the real spread along world X (24 mm against 127 mm). Real bundles stagger

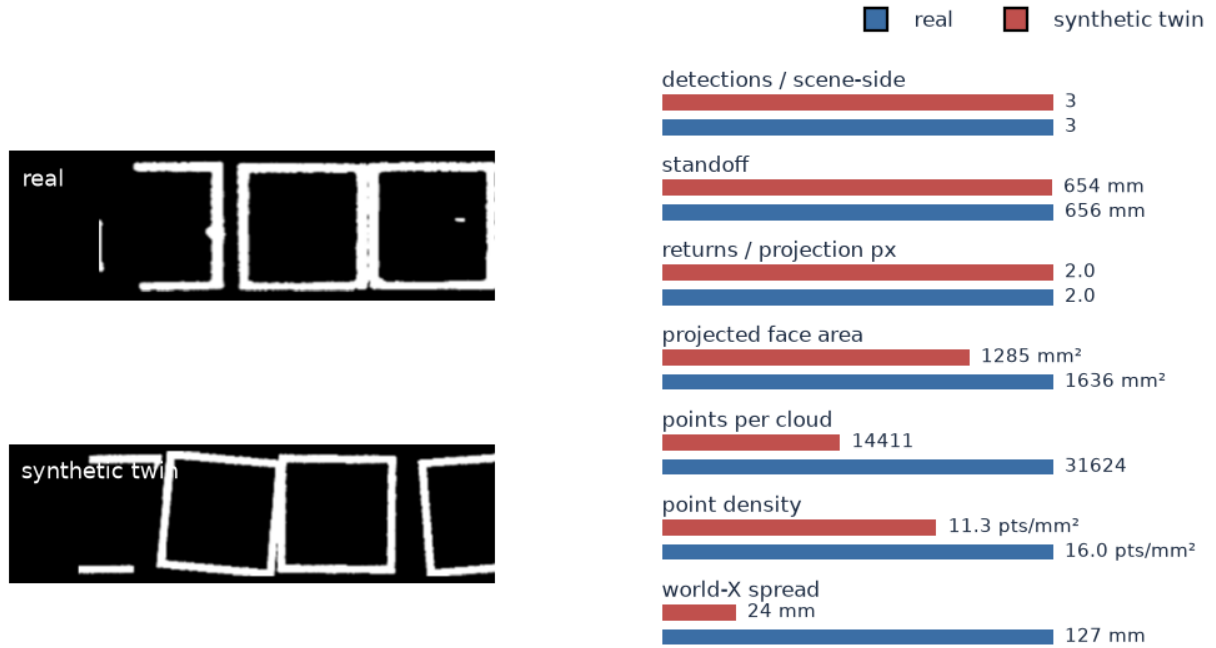


Figure 2.5: Twin against real. The detector’s input for a real scene of the square profile and for its twin, a simulated scene with the same profile and the same number of bars, real above and simulated below. On the right, the real and twin medians for seven quantities. Each row uses its own scale.

the ends of their bars along the bar axis, whereas those of the twin are nearly flush.

Consequently, the twin is a stand-in for relative comparisons only and no conclusion that depends on the world-X channel is drawn from it. Every result obtained on the twin remains in the YZ projection and in the cost units of the model itself.

Chapter 3

Identifiability of the carriage geometry

The available offline calibration is valid in the YZ projection plane but incorrect along the X-axis, i.e., along the bar direction. Reconstructing the length of a bar by connecting the centroids of its two end faces yields a value systematically smaller than the nominal value of 6005 mm. The median residual is -1077.47 mm, with a very narrow IQR, $[-1080.36, -1076.02]$. The residual is therefore an almost constant offset rather than random noise. The default length tolerance of the original system's pairing gate is ± 10 mm and none of the 425 pairs considered falls within it. This occurs because the carriage positions, which the controller reads at scan time, are not stored in the scan data available to this work. The offline calibration has a single knot per scanner, valid for one carriage position, so applying it evaluates the calibration at that position rather than at the actual one and introduces an error of about one metre along X.

While scene content alone recovers part of this geometry, it provably cannot recover the rest. The only metric datum that the recorded data preserves regarding a bar is its length, whose prior constrains only a single combination of the two carriage positions, leaving the orthogonal one free.

3.1 The length prior and its residual

Each bar is cut to a nominal length $L = 6005$ mm, so each pair of end faces, one per view, should reconstruct exactly that value.

Let s_1 and s_2 denote the positions of the two carriages along their respective translation axes and $w_1(s_1)$ and $w_2(s_2)$ the centroids in the world frame obtained by applying the calibration to those positions. The two views are also slightly offset in the projection plane. That offset is constant across the entire dataset along Z and varies scene by scene along Y (section 3.4), so the estimation involves a single unknown t_z and no global term along Y. For a pair k the residual is

$$r_k(s_1, s_2) = \left\| w_1^k(s_1) - \left(w_2^k(s_2) + (0, 0, t_z) \right) \right\| - L \quad (3.1)$$

The estimation of s_1 and s_2 minimises the median of the absolute value of the residuals:

$$(\hat{s}_1, \hat{s}_2) = \arg \min_{s_1, s_2} \text{median}_k |r_k(s_1, s_2)| \quad (3.2)$$

rather than the sum of squares because the pairs used for fitting include some uncertain associations whose large residuals could dominate a least-squares fit. The median absolute residual is far less sensitive to such outliers, making it the preferred choice here.

Since the objective is piecewise linear and exhibits non-differentiable points, the optimiser first performs a coarse grid search within the calibrated interval and then a local refinement, so that the result does not depend on an arbitrary initial point.

3.2 Identifiability of the carriage positions

The residual depends on the two carriage positions but not on each one independently. Owing to the structure of the model, if both carriages shift by the same amount along X , both reconstructed positions move together, so the distance between them stays the same. Every reconstructed bar length is therefore unchanged, as is every residual. A prior on the length therefore captures only the difference $s_1 - s_2$. The orthogonal direction, along which the two carriages slide together, belongs to the exact kernel of the Jacobian of the residual, so it is not a poorly observed direction that more data could improve. Since the observations carry no information about this direction, it cannot be recovered with more data, another optimiser or a different selection of scenes.

The degree to which the observed direction $s_1 - s_2$ is determined, by contrast, depends on the observations: on the number of pairs, their accuracy and their distribution. It is quantified by the curvature of the objective along that direction. The evaluation tool is the Gauss–Newton normal matrix $J^\top J$ of the residual with respect to the two parameters, divided by the number of pairs entering the fit to enable comparison across scenes of different sizes. It is the same observability analysis that the self-calibration literature applies to the parameters of a system Maye et al. 2016 [7], Nobre et al. 2019 [8], here reduced to a static problem with only two parameters.

Across all 425 pairs, the following eigenvalues are obtained

$$\lambda_1 = 2.008, \quad \lambda_2 = 2.91 \times 10^{-8}, \quad \lambda_1/\lambda_2 = 6.90 \times 10^7$$

The eigenvector associated with the largest eigenvalue, i.e., with the direction best determined by the data, is $(+0.708, -0.707)$, which corresponds to the direction $s_1 - s_2$ up to rounding errors. The matrix is obtained by linearising the residual around the estimate to which the alternation between calibration and assignment converges. Although linearisation is typically a local approximation whose result might depend on the evaluation point, this is not the case here because the null direction is exact across the entire parameter space and not only near the estimate. Consistently with this geometry, the second eigenvalue is zero within numerical precision.

Consequently, the objective is flat along that direction, so it has no interior minimum there and the search continues until it reaches the boundary of the calibrated domain. Whatever pair of individual positions an estimate reports depends on where that boundary lies and only the difference between the two is a measurement. Figure 3.1 shows this flat direction.

The same matrix also shows that no scene is geometrically more favourable than the others. The contribution of each pair to the curvature along the observed direction varies between

2.007 and 2.009 across all 108 scenes, i.e., it is practically constant. Each pair therefore carries almost the same information regarding the separation because the derivative of the distance between the two centroids with respect to a carriage position is close to 1 for any bar, regardless of its position, its inclination and the number of bars in the scene. The accuracy of the estimate depends on the number of pairs and their correctness, rather than the choice of favourable scenes.

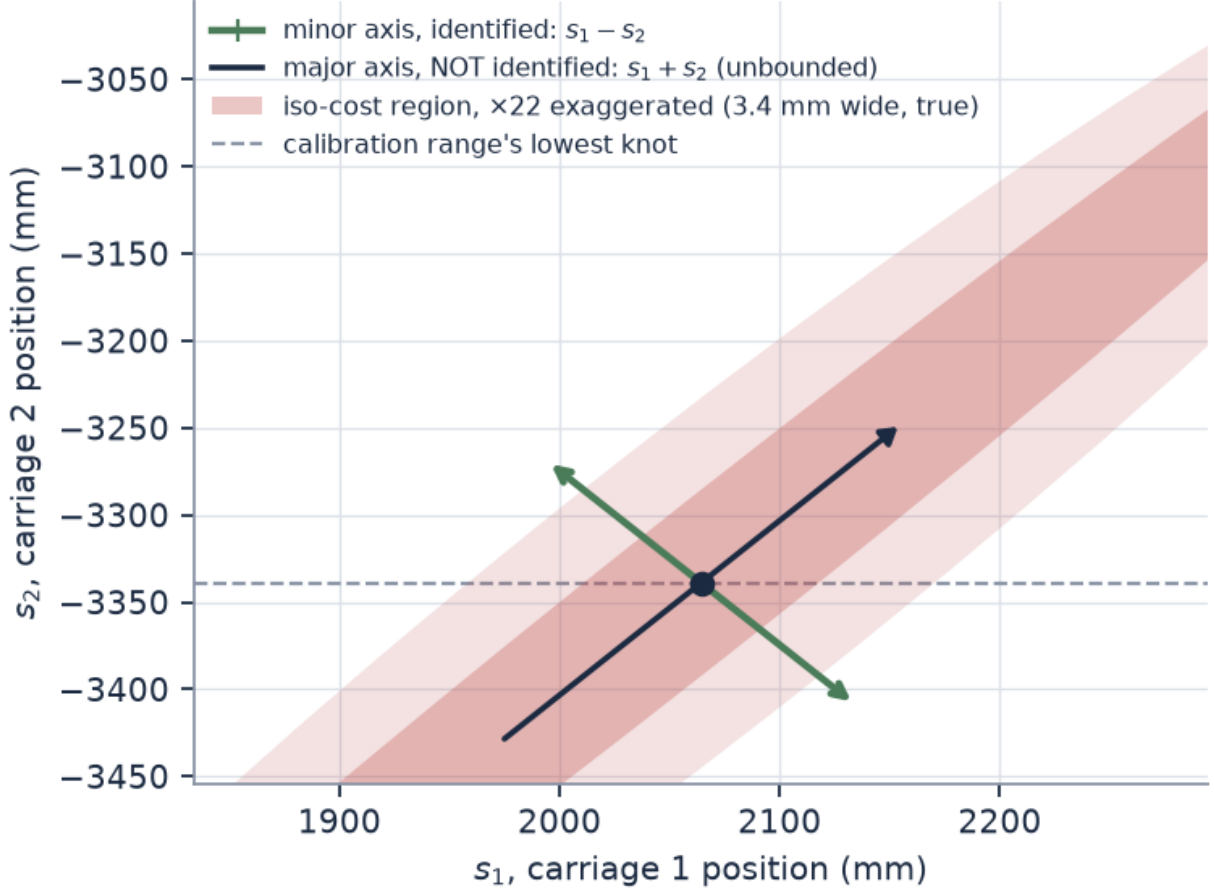


Figure 3.1: Unobservable direction of the length prior. Contour lines of the quadratic form in the two carriage positions s_1 and s_2 , plotted starting from the two eigenvalues and the eigenvector reported by the observability analysis. The cost grows rapidly along the $s_1 - s_2$ direction (the observed one), while remaining flat along the orthogonal direction in which the two carriages slide together. With a curvature ratio of 6.90×10^7 , the curves extend without bound in the flat direction. The width of the plotted strip is an empirical dispersion level, 3.4 mm at real scale, magnified 22 times to render it visible. The dashed line marks the lowest knot of the calibration interval, i.e., the boundary at which a search along the flat direction stops.

3.3 Recovering the carriage separation

The estimation requires pairs as input, yet these derive from an assignment that itself depends on the calibration. Calibration and assignment are therefore estimated alternately:

1. assignment is performed, under the current calibration;
2. the positions $\hat{s}_1, \hat{s}_2$ and the residual offset are re-estimated on the obtained pairs;

3. the process is repeated until the assignment is identical between two consecutive iterations.

Here convergence means that the assignment has stopped changing, not that a numerical tolerance has been reached. Two iterations are needed.

As the unobserved direction predicts, $\hat{s}_2$ settles at -3339.0 mm, the lower knot of the calibration interval, rather than at an interior minimum. Since no data constrain that direction, this value only marks where the search stops, whereas the difference $\hat{s}_1 - \hat{s}_2$ is a genuine measurement.

Table 3.1: Bar length residual under the three calibrations. The residual is the reconstructed bar length minus the nominal 6005 mm. For the offline calibration, a one-shot fit on a fixed assignment and the joint self-calibration, the table reports the median residual, its interquartile range in millimetres and the number of pairs within the default ± 10 mm tolerance of the length check of the original system. The first and last rows use the pairs of the hybrid method, whereas the one-shot fit uses those of free assignment.

bar length residual	median	IQR	within ± 10 mm
offline calibration with single knot	-1077.47 mm	$[-1080.36, -1076.02]$	0 / 425
one-shot fit on fixed assignment	-1.48 mm	$[-5.11, +0.15]$	359 / 415
joint self-calibration	-0.15 mm	$[-2.99, +1.38]$	403 / 425

The first and last rows of table 3.1 use the hybrid method, on the offline calibration and on the recovered one, respectively. The middle row instead uses a free assignment and keeps the pairs fixed throughout the process. It therefore differs from the joint fit in two ways: the origin of the pairs and their ability to change. Its median is nominally closer to zero, but this is uninformative given that the median is precisely the quantity optimised by the objective. The informative columns are the dispersion and the fraction of pairs within ± 10 mm. With an IQR of 4.36 mm against 5.26 mm the joint fit is tighter, as expected, because it no longer fits the calibration to pairs that the calibration itself would have discarded.

The recovered separation is

$$\hat{s}_1 - \hat{s}_2 = 5403.3 \text{ mm}$$

and has an independent verification. Both the replay tool and the test harness of the original system use a default carriage position of 2701 mm per side, i.e., 5402 mm of separation, described in its documentation as a dashboard convention. The estimated value, obtained solely from scene content and the length prior without access to that constant, agrees with it within 1.3 mm.

None of this establishes that the recovered separation is accurate in an absolute sense. No independent metrology of this rig exists and the constant of the original system, however close, is a software convention rather than a measured distance. The fit also has two free parameters and only one identifiable combination. A richer calibration model, with per-knot corrections or a rotational term, is not tested here and is not guaranteed to remain identifiable from the same prior.

Regarding uncertainty, the quantity to quote is the reproducibility across scenes rather than the repeatability across pairs. Estimating the separation independently on each scene yields a median of 5403.0 mm and an IQR of $[5401.9, 5405.3]$. A bootstrap on the joint fit

that resamples bundles (section 2.1) gives [5402.8, 5403.8]. Figure 3.2 shows the separation estimated on each scene.

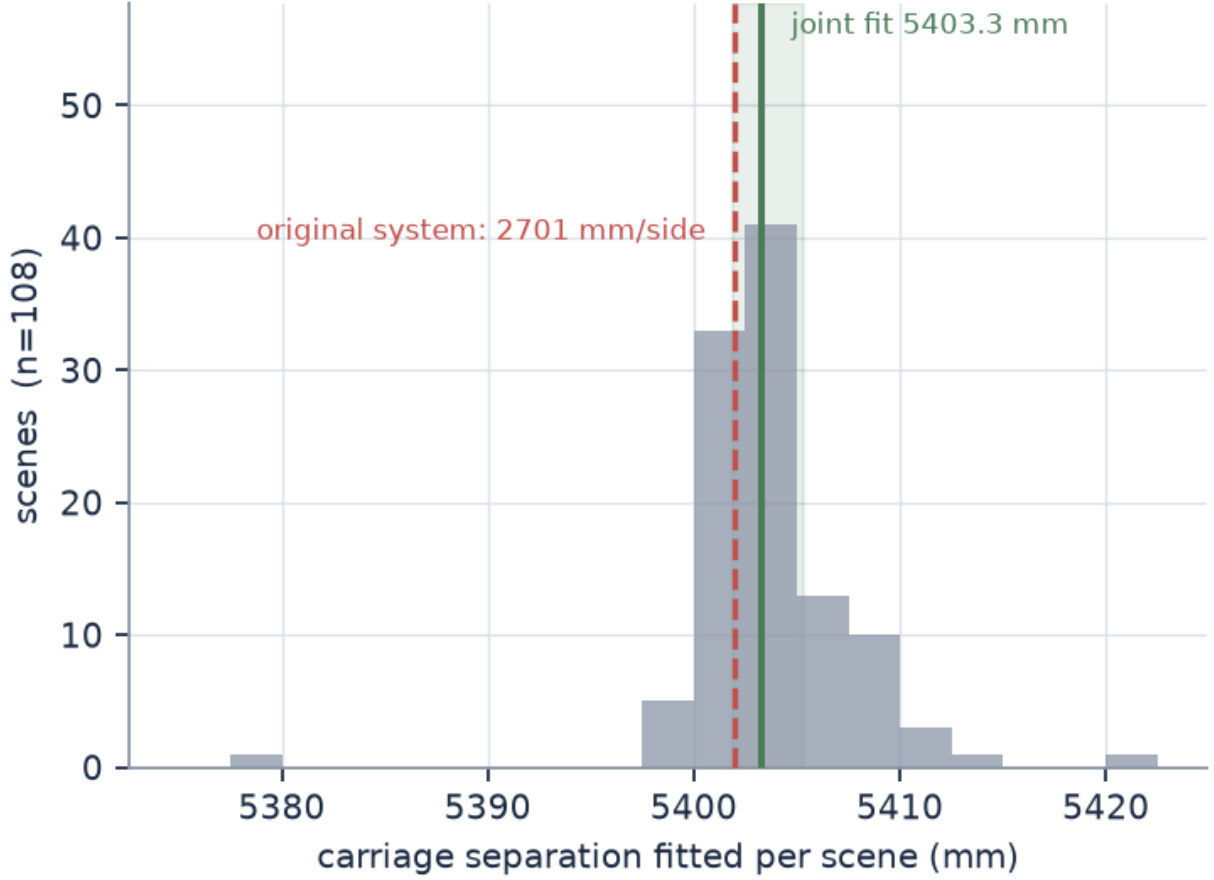


Figure 3.2: Cross-scene reproducibility of the carriage separation. The carriage separation estimated independently on each of the 108 scenes. The joint fit of 5403.3 mm is shown with the interquartile range of the per-scene values, along with the working constant of the original system (2701 mm per side), a value to which the fit never had access.

The dispersion across scenes indicates matching quality rather than calibration quality. The outliers comprise one scene with two pairs at one extreme (5378.2 mm) and one with four pairs at the other (5421.2 mm). A scene whose pairs are wrong will estimate whatever separation makes those pairs consistent, regardless of how many it contains.

3.4 Calibration in Z, transport in Y

A residual offset between views in the projection plane could stem from a miscalibrated axis or a transport of the bar between the two acquisitions. The former is constant, whereas the latter varies from scene to scene because the two views of a scene are acquired at different times while the conveyor moves. A variable offset therefore cannot be attributed to calibration.

The median cross-view offsets per scene, computed over 108 scenes for each axis, are reported below. Each scene's offset is the median across its pairs.

- Z: median +1.64 mm, IQR [+1.12, +2.30]. With an IQR width of 1.18 mm, the Z offset varies by little more than a millimetre across scenes, a magnitude compatible with a constant calibration offset.

- Y: median -5.36 mm, IQR $[-47.56, +21.04]$. The IQR width is 68.60 mm, nearly sixty times that of Z, while 62 of the 108 scenes exhibit an offset exceeding ± 30 mm. The dispersion is thus wider and heavy-tailed, with the variability concentrated along the transport axis.

Consequently, the Z component is attributed to calibration, while the Y component is attributed to transport. This is why the residual described in section 3.1 includes a dataset-wide constant term along the Z-axis, but none along the Y-axis. This distinction forms the basis of a choice already adopted by the pairing methods: aligned assignment, monotonic alignment and the hybrid method all estimate a scene-specific YZ translation via RANSAC consensus, rather than relying on calibration along the Y-axis.

3.5 Metric recovery and the combinatorial ambiguity

Recalibration does not alter the relative performance of the pairing methods. The counting F1 varies by at most 0.002 for each of the methods re-scored. The exception is the order-based anchor with the length check of the original system, which jumps from 0.000 to 0.917. This jump comes from removing the -1077 mm bias, which made every pair fail the check. Once the bias is removed the check can be passed again, although this does not mean that it separates correct pairs from incorrect ones (section 4.1).

None of the results presented in this chapter addresses the ambiguity that forms the core subject of this thesis. Recovering the separation solves the metric problem of reconstructing a cross-view pair with the correct length. The combinatorial problem remains unchanged, as a bar reconstructed with the correct length provides no indication of which among several identical neighbouring elements was actually paired with it. This is the first of the properties of the shift family stated in section 2.5: under exact regularity, geometry does not distinguish a shifted correspondence from the true one. The metric recovery in this chapter does not affect that property because it operates on the calibration rather than the combinatorial ambiguity.

Chapter 4

Shift ambiguity after self-calibration

Self-calibration removes the metric error of the rig, so that a correct cross-view pair reconstructs the nominal bar length (section 3.3). The length of a pair, however, gives no indication of which of several identical neighbouring bars a face belongs to. This combinatorial ambiguity is described by the shift family π_k of section 2.5, which pairs each face on one side with the face k positions further along on the other.

4.1 Sensitivity of the length check to a one-bar shift

The aggregation gate of the original system accepts a candidate pair only when the distance between the centroids of its two end faces lies within ± 10 mm of the nominal bar length L (section 2.4). Of the three checks in this gate, it is the only one based on a property of the product itself, whereas the torsion and tilt checks, like every term of the shared cost model, compare the geometry of the two faces.

Consider a member π_k of the shift family, which pairs the end of bar i on one side with the end of bar $i + k$ on the other. The reconstructed segment then joins two different bars and becomes the hypotenuse of a right-angled triangle whose legs are the bar length L and the lateral offset kp , where p is the bar pitch. The reconstructed length therefore changes by

$$\Delta\ell(k) = \sqrt{L^2 + (kp)^2} - L \approx \frac{(kp)^2}{2L} \quad (4.1)$$

with $L = 6005$ mm, so the change grows with the square of the offset. Table 4.1 gives $\Delta\ell$ for a one-bar shift at the measured pitch of each family (fig. 2.4), in millimetres and in units of $\sigma = 3.23$ mm, the standard deviation that a normal distribution would need to reproduce the interquartile range of the length residual after self-calibration (section 3.3).

Since the pitch is much smaller than the bar length, a one-bar shift changes the reconstructed length by at most about 1 mm, against a tolerance of ± 10 mm.

Setting $\Delta\ell = 10$ mm in the formula gives the lateral offset at which the check would react, $kp \approx 347$ mm, which is about 9 pitches for the square profile, 5 for the L-profile and 3 for the complex profile, the family with the largest pitch. More than nine clouds in ten have a narrower scan window than this (median 258 mm, 90th percentile 337 mm), so only

Table 4.1: Length change caused by a one-bar shift. For each family, the measured bar pitch, the change $\Delta\ell$ in the reconstructed length when a pair joins the ends of two adjacent bars and the same change in units of $\sigma = 3.23$ mm, the standard deviation of a normal distribution with the same interquartile range as the length residual after self-calibration. The length check of the original system has a tolerance of ± 10 mm.

family	pitch (mm)	$\Delta\ell$, one pitch (mm)	$\Delta\ell/\sigma$
L-profile, second acquisition	38.6	0.12	0.04
square profile	40.2	0.13	0.04
L-profile	69.2	0.40	0.12
heavy profile	96.0	0.77	0.24
complex profile	118.4	1.17	0.36

the widest ones, up to 414 mm (section 2.2), could contain such an offset. A threshold of $3\sigma \approx 9.7$ mm in place of the fixed ± 10 mm gives almost the same offset (341 mm), so the limit lies in the geometry of the check and not in the choice of tolerance.

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